A 4-year study of hamstring injury outcomes in elite track and field using the British Athletics rehabilitation approach

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ABSTRACT

Objectives The British Athletics Muscle Injury Classification (BAMIC) correlates with return to play in muscle injury. The aim of this study was to examine hamstring injury diagnoses and outcomes within elite track and field athletes following implementation of the British Athletics hamstring rehabilitation approach.

Methods All hamstring injuries sustained by elite track and field athletes on the British Athletics World Class Programme between December 2015 and November 2019 that underwent an MRI and had British Athletics medical team prescribed rehabilitation were included. Athlete demographics and specific injury details, including mechanism of injury, self-reported gait phase, MRI characteristics and time to return to full training (TRFT) were contemporaneously recorded.

Results 70 hamstring injuries in 46 athletes (24 women and 22 men, 24.6±3.7 years) were included. BAMIC grade and the intratendon c classification correlated with increased TRFT. Mean TRFT was 18.6 days for the entire cohort. Mean TRFT for intratendon classifications was 34±7 days (2c) and 48±17 days (3c). The overall reinjury rate was 2.9% and no reinjuries were sustained in the intratendon classifications. MRI variables of length and cross-sectional (CSA) area of muscle oedema, CSA of tendon injury and loss of tendon tension were associated with TRFT. Longitudinal length of tendon injury, in the intratendon classes, was not associated with TRFT.

Conclusion The application of BAMIC to inform hamstring rehabilitation in British Athletics results in low reinjury rates and favourable TRFT following hamstring injury. The key MRI variables associated with longer recovery are length and CSA of muscle oedema, CSA of tendon injury and loss of tendon tension.

INTRODUCTION

Hamstring injury is common in sports that require sprinting and acceleration.[1–5] The rate of hamstring injury is increasing in elite football and baseball.[1 3] Reported recurrence rates in prospective follow-up studies range from 14% to 63%.[3 6 7] In track and field, hamstring injury is the most prevalent injury in competition and therefore has significant performance and financial implications for athletes, national governing bodies and international federations.[8–10]

The British Athletics Muscle Injury Classification (BAMIC) is an MRI classification system with clearly defined, anatomically focused classes based on the site of injury: myofascial (class a), muscle–tendon junction (class b) or intratendon injury (class c) and a numerical grading system (0–4) based on the extent of injury (table 1).[11] It is a reliable classification system that is associated with return to play.[7 12–18] A retrospective injury review in British Athletics between 2010 and 2014, reported an increase in time to return to full training (TRFT) and reinjury rate in injuries that extended into the hamstring intramuscular tendon (class c).[7] An extended TRFT or increased hamstring reinjury rate in injuries that involve the intramuscular tendon has also been seen in other studies[19–24] although a prospective study, predominately in footballers, has questioned the clinical relevance of injury to the intramuscular tendon in some athlete groups.[23 25] The muscle tendon unit demands in a return to elite sprinting are likely to be significantly increased relative to field sports, such as football, particularly at subelite levels, where maximal speed is less and players may reduce their running speed and distance but still return to play.[26 27]

A small number of hamstring rehabilitation programmes have been published.[28–35] As individual muscles have different architecture and function[36 37] and there are different prognostic outcomes in certain structural injuries,[7 24 38 39] a targeted rehabilitation programme based on the specific injury appears prudent. This is the clinically reasoned, published, rehabilitation approach advocated by the medical team at British Athletics, the national governing body supporting elite British track and field athletes.[28] The rehabilitation process is informed by the BAMIC classification and the supporting principle that different tissues within the hamstring undergo different healing processes.[40–42]

The aim of this paper is to report hamstring injury diagnoses and outcomes between 2015 and 2019 in an elite track and field programme following the implementation of the BAMIC and the associated targeted rehabilitation approach.

METHODS

Inclusion criteria

The British Athletics Medical team provides full time medical care to track and field athletes on the elite Olympic World Class Programme (WCP). Entry to the WCP is by the judgement of a panel of high performance coaches, who deem the athlete to have the potential to win an Olympic or World Championship medal in track and field. All injury episodes were recorded on the Smartabase electronic medical record (EMR). Athletes were eligible for inclusion in this prospective study if they;

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Table 1 Summary of British Athletics Muscle Injury Classification system

Grade	Description	MRI
0	Focal or generalised muscle pain following exercise	Normal or patchy high signal change throughout one or more muscles.
1a	Small myofascial tear	High signal change evident at the fascial border with less than 10% extension into muscle belly Craniocaudal distance of < 5 cm.
1b	Small muscular/muscle–tendon junction tear	High signal change of less than 10% cross-sectional area of muscle usually at the muscle-tendon junction High signal change of craniocaudal length <5 cm (may note fibre disruption of <1 cm)
2a	Moderate myofascial tear	High signal change evident at fascial border with extension into the muscle High signal change cross-sectional area of between 10% and 50% at maximal site High signal change of craniocaudal length >5 cm and <15 cm Architectural fibre disruption usually noted over less than 5 cm
2b	Moderate muscle–tendon junction tear	High signal change evident within the muscle, usually at the muscle-tendon junction High signal change cross-sectional area of between 10% and 50% at maximal site High signal change of craniocaudal length >5 cm and <15 cm Architectural fibre disruption usually noted over less than 5 cm
2c	Moderate sized intratendinous tear	High signal change extends into the tendon with longitudinal length of tendon involvement <5 cm Cross-sectional area of tendon involvement <50% of maximal tendon cross-sectional area No loss of tension or discontinuity within the tendon
3a	Extensive myofascial tear	High signal change evident at fascial border with extension into the muscle High signal change cross-sectional area of greater than 50% at maximal site High signal change of craniocaudal length of greater than 15 cm Architectural fibre disruption usually noted over more than 5 cm
3b	Extensive muscle–tendon junction tear	High signal change cross-sectional area of greater than 50% at maximal site High signal change of craniocaudal length of greater than 15 cm Architectural fibre disruption usually noted over more than 5 cm
3c	Extensive intratendinous tear	High signal change extends into the tendon Longitudinal length of tendon involvement >5 cm Cross-sectional area of tendon involvement >50% of maximal tendon cross-sectional area There may be loss of tendon tension although no discontinuity is evident
4	Full thickness tear of muscle	Complete discontinuity of the muscle with retraction
4c	Full thickness tear of tendon	Complete discontinuity of the tendon with retraction

Modified with permission from Pollock *et al*.[11]
If any characteristics of a higher grade injury are present the injury is graded at the highest grade.

reported an injury to the posterior thigh during or immediately after a training session or competition between December 2015 and November 2019; the injury was subsequently assessed and the athlete referred for an MRI scan within 7 days of the injury; and rehabilitation was prescribed by the British Athletics medical team.

Injury characteristics

The athlete's gender, age, athletic discipline (Short Sprint, Long Sprint, Other Power, Endurance) and date of injury were recorded. Injury mechanism and other injury specific variables: training venue, mode of onset, self-reported gait phase, coach-reported training intensity and MRI characteristics were also recorded in a bespoke hamstring injury EMR.

MRI protocol

Each MRI scan was assessed by the same specialist musculoskeletal radiologist and specific injury characteristics were recorded. The injury was classified using BAMIC (table 1). MRI scans were performed on a 3 T MRI scanner at one of three imaging centres in the UK. All MRI scans included a combination of axial, sagittal and coronal T1-weighted, T2-weighted fat suppressed/proton density-weighted fat suppressed or short T1 inversion recovery (STIR) sequences.[43] MRI analysis was performed on McKesson Picture Archive Communication System 64-bit professional workstation. The freehand region of interest tool was used to assess cross-sectional area (CSA) of muscle oedema (high signal change on T2/STIR sequences) and overall muscle area. The same tool was used to assess CSA of tendon abnormal signal versus overall tendon cross section. The length of oedema within the muscle or within the tendon was measured on coronal and/or sagittal images using the distance measuring tool. Loss of tension was defined as lack of taut tendon appearance within the muscle on coronal and/or sagittal images, producing the 'wavy tendon' sign.

Rehabilitation approach

The rehabilitation programme followed the principles of the British Athletics hamstring injury management approach.[28] This is a functional progressive rehabilitation programme that is informed by the nature of the structural injury, the athlete's functional requirements and a shared decision-making process with the coach and athlete to determine the performance goals. The rehabilitation was provided by full time British Athletics doctors and physiotherapists. If athletes were based abroad, British Athletics' doctors and physiotherapists would collaborate with a local practitioner to structure the rehabilitation to these principles. The principles of the British Athletics rehabilitation approach are summarised in box 1 and figure 1. While exercise prescription is key, the rehabilitation approach is also

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Box 1 Management principles of hamstring injury rehabilitation in British Athletics[28]

1. Establish an accurate structural diagnosis and British Athletics Muscle Injury Classification (BAMIC) injury classification
2. Facilitate the collaborative expertise of the sports science and medicine team
3. Involve the coach and athlete in shared decision making
4. Train movements and muscles
5. Prescribe strength exercises to achieve a specific goal
   - *Develop high eccentric force*
   - *Increase fascicle length to enhance length–tension relationship*
   - *Develop muscle–tendon unit specificity*
   - *Develop fatigue resistance*
   - *Overcome selective muscle inhibition*
6. Apply an individualised non-reductionist model targeting contributing risk factors

characterised by other management principles such as shared decision-making with athlete and coach, risk factor modification and multidiscipline expertise. With respect to exercise prescription, myofascial injuries (class a) are characterised by a functional and expedited return to training, where running based activities are the focus. Muscle–tendon junction injures (class b) are characterised by progressive strengthening and running-based activities guided by the achievement of objective functional milestones. Intratendon injuries (class c) are characterised by a more conservative approach initially, where early lengthening contractions on the tendon are avoided, to respect slower tendon healing.

Outcome measures

Injuries were evaluated by gender, location and occurrence (training/competition and self-reported gait phase).

The TRFT, defined as completing unrestricted sprint efforts at full pace in spikes, was recorded in the athlete's EMR during the clinical rehabilitation.

Repeat injury to the same hamstring muscle during rehabilitation (exacerbation) or within 3 months of return to full training (recurrence) was recorded.[44 45] A recurrence or exacerbation was recorded if the athlete sustained an acute, sudden return of hamstring pain during exercise, followed by deterioration of functional and clinical tests that required cessation of current activity and subsequent modification of rehabilitation or training for greater than 48 hours.

Statistical analysis

The relationships between dependent (TRFT) and explanatory (BAMIC, grade and location of the injury, muscle injured, MRI observations, activity and self reported gait phase at time of injury) variables were tested by Kruskal Wallis, with Dunn's post-hoc where applicable, or Fisher's exact test. If athletes suffered a repeat injury during rehabilitation (exacerbation), this injury was excluded from the TRFT analysis but recorded as a reinjury. The relationships between TRFT and the grade and sites (class a–c) of the injuries were examined using linear regression analysis controlled for age and gender. A custom Python script was used to conduct the statistical analysis, with the significance threshold set at $p \leq 0.05$.

RESULTS

Seventy hamstring injuries in 46 athletes (24 women and 22 men, 24.6±3.7 years) were included in the study. Eighty-seven per cent of injuries occurred in the Sprint/Power group (Short Sprints: 40%, Long Sprints: 25%, Other Power: 21%) and 13% of injuries occurred in the Endurance athlete group. Eighteen athletes sustained more than 1 independent hamstring injury during the study period (15 athletes (two injuries), 2 athletes (3 injuries) and 1 athlete (4 injuries)). In athletes with more than 1 recorded injury during the study, 4 of these were within 1 year and on the same side as the initial injury, but none were at the same location or within the same muscle (grade 2b semitendinosis (ST) followed by a grade 0 injury at 5 months; grade 2a ST followed by grade 0 injury at 4 months, grade 0 followed by grade 1b biceps femoris long head (BFLH) (7 months) and a grade 0 injury followed by a grade 0 injury at 5 months). Only 46% of injuries occurred at the athlete's usual training venue. Injuries otherwise occurred at different UK training tracks, international training camps or in competition.

Injury location

Most injuries occurred in the distal third of the hamstring (43%), with 31% in the proximal third and 26% in the central third. An isolated injury to the BFLH muscle was the most frequently occurring structural muscle injury (70%), followed by injury to the semi-membranosis (19%), multiple muscles (6%), semitendinosis (4%) and the short head of biceps femoris (2%). There were 16 grade 0 injuries with a normal MRI scan. The characteristics and location of the injuries are included in tables 2 and 3).

Injury occurrence

Seventy-one per cent of injuries occurred during training, 4% during competition warm up and 24% during competition. The coach-reported specific athletic activity undertaken during the injury was sprinting (>90% maximal velocity) in 54% and running at between 50% and 90% maximal velocity in 27%. Injuries occurred during jumping activities in 8% of cases. Athletes self-reported the phase of the running stride at which they felt the injury occur in 89% of injuries. The single most self-reported phase at injury was during the stance phase (37%), other (including unsure, relay change, finishing line) was reported in 26%, terminal swing (18%) and a gradual subacute presentation after the activity in 19% of cases.

TRFT: BAMIC and TRFT

The median TRFT for the different BAMIC hamstring injuries is described in table 4 and figure 2. There was a significantly shorter TRFT for grade 0 injuries with respect to every other grade (grade 0 vs grade 1, $p=0.03$; grade 0 vs grade 2 $p<0.001$; grade 0 vs grade 3 $p<0.001$). There was also a significant difference between grade 1 vs grade 2 injuries ($p=0.03$) and grade 1 and grade 3 ($p=0.004$). There was no difference in TRFT between grade 2 and grade 3 injuries ($p=0.14$). There was a highly significant difference in TRFT between class c (intratendon) injuries versus class a (myofascial, $p=0.0009$) and class c versus class b (muscular or muscle–tendon junction, $p=0.0009$) injuries. There was no difference in TRFT between classes a and b injuries ($p=0.32$).

MRI variables and TRFT

In the structural (MRI positive) injuries, the length of muscular oedema ($p=0.0001$) and CSA of muscular oedema

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Figure 1 Summary of hamstring rehabilitation within British Athletics. Reproduced with permission from MacDonald *et al.*[28]

(p=0.0005) were significantly associated with TRFT. The length of intratendon injury was not associated with TRFT (p=0.16) but the CSA of tendon injury was significantly associated with TRFT (p=0.003). Loss of tendon tension was associated with a longer TRFT (p=0.02). There was no significant difference in outcome if the injury site was the distal T junction of the long head and short head of biceps femoris (p=0.85). There was no difference in TRFT between proximal and distal injuries.

Activity and self-reported gait phase at time of injury and TRFT

Injuries that were sustained in competition took on average 28.7 days (±18.6) to return to full training compared with 17.7 days (±10.3) for those injuries sustained in training (p=0.09). Intratendon class c injuries occurred more frequently in competition than training (p=0.004). There was no significant difference in outcomes related to the self-reported phase of the running cycle at which the injury occurred or the training activity that was being undertaken at the time of injury.

Reinjuries

Only two repeat injuries (recurrence or exacerbation) were sustained in this cohort of 70 hamstring injuries over a 4-year period: in one index 1a and one index 2b injury. This gives a reinjury rate of 2.9% in this elite athlete group.

DISCUSSION

The main finding of this paper is that implementing the BAMIC rehabilitation approach to hamstring injuries was

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Table 2 Characteristics of the study participants and the hamstring injuries

Variable	Number	Per cent
Study participants	47	
Gender		
Male	22	47
Female	24	53
Median age in years (IQR, range)	24 (4, 19–37)	
Injuries	70	
Grade 0 injuries	16	23
Median TRFT in days (IQR, range)	17 (13, 4–70)	
Location of injury	n=54	
Proximal	17	31
Central	14	26
Distal	23	43
Muscle injured	n=54	
Biceps femoris long head	38	70
Semi-membranosis	10	19
Semi-tendinosis	2	4
Biceps femoris short head	1	2
Multiple	3	6
Reinjury (recurrence or exacerbation)	2 (1a and 2b)	3
Injured during training or competition	n=70	
Training	50	71
Competition	17	24
Warm up for competition	3	4
Activity during which injury occurred	n=59	
Sprinting (>90% maximum velocity)	32	54
Running (50%–90% maximum velocity)	16	27
Jumping	5	8
Other	6	10
Self-reported phase of gait at injury	n=62	
Stance	23	33
Other/unsure	16	23
Gradual subacute	12	17
Terminal swing	11	16

TRFT, time to return to full training.

associated with a very low reinjury rate in this elite athlete cohort. A total reinjury rate of 2.9%, and no reinjuries within the intratendon classification, compares favourably to previous work in elite track and field athletes and other elite sports where reinjury rates are usually between 12% and 25%, and at times higher in injuries that involve the tendon.[7 24 25 46–48] Similarly low reinjury rates have been reported.[34 49] Askling *et al* compared a lengthening rehabilitation protocol to what was described as a conventional rehabilitation protocol.[49] Two reinjuries (n=56) were reported

Table 4 Time to return to full training in the different British Athletics Muscle Injury Classification (BAMIC) classes

BAMIC	Number of injuries	Median TRFT (days (IQR, range))
0	16	9.0 (4.3, 4–17)
1a	6	12.0 (4.0, 9–16)
1b	19	17.0 (10.3, 8–32)
2a	4	19.0 (3.5, 14–21)
2b	11	19.0 (3.5, 11–36)
2c	7	35.0 (9.5, 25–43)
3a	1	17.0
3c	6	51.5 (23.8, 28–70)

TRFT, time to return to full training.

in the conventional protocol. However, the average mean time to return of 49 days (lengthening protocol) and 86 days (conventional protocol) was far greater than the mean return times in our study. Mendiguchia *et al* also reported a 4% reinjury rate using their multifactorial criteria-based progressive algorithm for hamstring rehabilitation, although the study population was limited to footballers with grade 1 muscle injuries.[34]

The BAMIC injury grade and the intratendon class c classification are associated with longer TRFT in elite track and field athletes when this rehabilitation approach is prescribed. This study supports previous work that muscle injuries involving the tendon have a longer TRFT but can have an excellent outcome without surgical intervention.[7 19–22 24 25] While performance and rehabilitation relevance may vary between different sports, tendon tissue heals more slowly and differently to muscle, requiring collagen synthesis and remodelling within an extracellular matrix scaffold.[40 50] The tendon has an important role in muscle–tendon unit force production and demand increases non-linearly as running speed increases.[27] The British Athletics rehabilitation approach for tendon injuries is detailed in previous work.[28] The loading principles are: an initial longer period of protection for the healing tissue by avoiding eccentric exercises or activities with high 'elongation stress'[51]; early isometric loading to facilitate tendon adaptation, progressing to eccentric loads at gradually increasing lengths; and tendon specific loading considerations, including heavy and long-term loading. It is also important to note that the rehabilitation approach is not merely an exercise protocol. A collaborative, informed multidisciplinary team and shared decision-making with athlete and coach are critical principles within the British Athletics rehabilitation approach which enhance communication and sharing of knowledge.[28]

Table 3 Location of injury in different British Athletics Muscle Injury Classification (BAMIC) categories

BAMIC	Total number of injuries	Proximal	Central	Distal	BLFH	SM	ST	Other/mixed
1a	6	0	3	3	5	1	0	0
1b	19	8	1	10	10	6	0	3
2a	4	1	1	2	3	0	1	0
2b	11	6	2	3	8	1	1	1
2c	7	2	2	3	7	0	0	0
3a	1	0	1	0	1	0	0	0
3c	6	1	4	1	4	2	0	0

BFLH, biceps femoris long head; SM, semi-membranosus; ST, semitendinosis.

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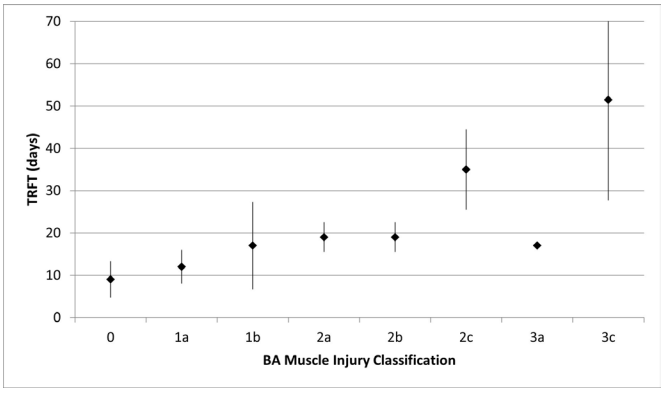

Figure 2 Median±IQR (days) time to return to full training (TRFT) in different British Athletics Muscle Injury Classification (BAMIC) categories.

This overall rehabilitation approach has not only resulted in very low recurrence rates but also favourable TRFT times (figure 2). The TRFT is lower in all but one injury category (the 2c category), relative to previous work in track and field. In a previous 4-year study from 2010 to 2014, the average TRFT for 1a injuries was 18±4 days; for 1b 18±11 days; for 2b 21±10 days; for 2c 27±7 days and for 3c injuries 84±49 days.[7] In this current study, the mean TRFT for the 2c category was longer by 7 days but all other categories had lower TRFT. Of particular note was the 3c class (figure 3) which had a mean TRFT of 48±17 days. These

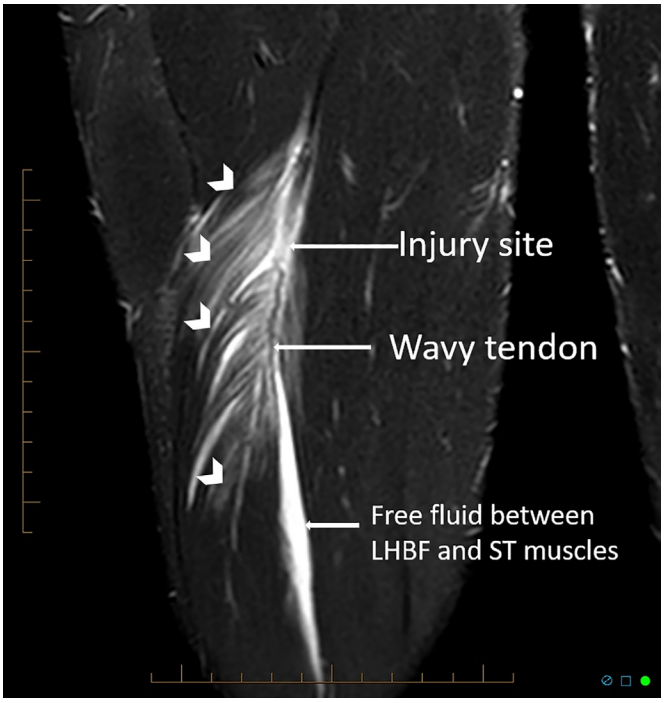

Figure 3 Coronal fat-saturated proton density weighted MRI of a male athlete with acute right mid-thigh hamstring injury. Typical feathery intramuscular oedema is seen surrounding an acute proximal third long head biceps femoris (LHBF) intramuscular tendon injury with loss of tension and wavy tendon sign on the distal side of the injury epicentre. The loss of tension is indicative of a British Athletics Muscle Injury Classification 3c injury. The injury has occurred below the division of the proximal conjoint long head biceps femoris and semitendinosis (ST) tendon.

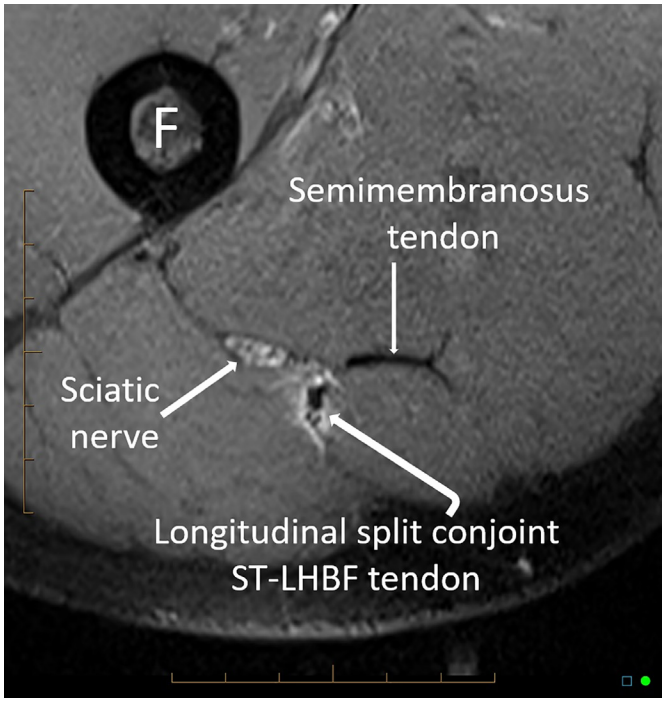

Figure 4 Axial fat-saturated proton density weighted MRI of a female athlete with acute proximal third right hamstring injury. There is a coronal-orientated split within the proximal conjoint long head biceps femoris (LHBF) and semitendinosis (ST) tendon (angled arrow). The split involved less than 50% of tendon cross section and extended over a length of 2 cm without loss of tendon tension. Therefore, the injury is classified as 2c by British Athletics Muscle Injury Classification criteria. Note the close relationships of the conjoint and semimembranosus tendons with the sciatic nerve (F=femoral shaft).

favourable TRFT times suggest that the low recurrence rates in the current study are not a result of an overall slower, cautious rehabilitation approach.

The BAMIC grade also correlated with TRFT with this rehabilitation approach. Correlation between BAMIC and RTP has also been found in a number of previous studies.[7 13 15–17] Consistent with previous research, grade 0 injuries have a short TRFT with low recurrence. Several radiological features, including length and CSA of muscle oedema, were associated with differences in TRFT. This suggests that these original criteria for BAMIC radiological grading were appropriate and do not need revision. However, only the CSA of tendon involvement and not the longitudinal length of tendon involvement were significantly associated with TRFT. This may be due to tendon tension and muscle–tendon unit integrity being maintained with intratendon longitudinal splits but disrupted with larger CSA tendon injuries. Long intratendon splits are often detected easily in multiple axial MRI sections (figure 4) and the intratendon involvement may extend for many centimetres. An update of the BAMIC system could consider removing or changing the length of tendon injury as a differentiator between classes 2c and 3c. The CSA of tendon injury and loss of tendon tension (waviness) appear to be key prognostic radiological findings in the intratendon class of injury. Consistent with previous work, 1a injuries appear to have a positive prognostic outcome. Further work is required to understand whether myofascial injuries warrant an independent classification and subsequent rehabilitation approach.

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All cases in this prospective study were managed with expert rehabilitation and without surgical intervention or platelet rich plasma (PRP) injection. The low reinjury rates and reasonable TRFT times suggest that non-surgical management is usually appropriate for 2c and 3c injuries to enable return to elite level sprinting. Further work to determine surgical criteria for high grade (particularly 4c) intratendon hamstring injuries is required.

The most common mechanism of injury reported was high speed sprinting. Unsurprisingly, the most commonly injured muscle was the long head of biceps, which previous research has determined undergoes the largest strain during maximal sprinting.[24 37 52–55] One method of describing hamstring injury mechanism is to classify into stretch and sprinting types.[21] Stretch mechanism injuries have been associated with more proximal injury to the semimembranosus, while sprinting mechanism is distributed between proximal and distal portions of the long head of bicep femoris.[56] Our data are consistent with this, as 43% of injuries occurred in the distal third of long head of bicep compared with 31% in the proximal third.

A potential explanation for the increased incidence of distal bicep injuries compared with other research[49] could be the interaction between architectural differences along the length of the hamstring muscle and aberrant sprinting biomechanics. The distal BFLH has shorter fascicles than the proximal BLFH (6.35 cm vs 7.12 cm) and the ST (6.35 cm vs 15.49 cm).[57] This makes it less well adapted for increased excursion of the tendon due to knee extension in the end swing phase.[58 59] Excessive knee extension, and trunk flexion, at end swing phase has been suggested as a potential biomechanical fault leading to hamstring injury, as it leads to excessive strain on the bicep.[59–61] If the BFLH is less well adapted to accommodate increased knee extension, this will be magnified during faulty running mechanics—such as over-striding—that causes a further increase in knee extension and consequently biomechanical load on the bicep.

Consistent with previous research in track and field most injuries occurred during training,[62] possibly due to greater exposure to high speed running and fatigue in training relative to competition. However, class c injuries occurred relatively more often during competition. As running velocity increases, tendon strain and work done increase in a non-linear manner.[27 63] An increased exposure to maximal velocity running in competition may account for the increased proportion of tendon injuries during competition.[64]

Strengths and limitations

The strength of this study was its prospective nature and the creation of a bespoke EMR to assist with consistent radiological and clinical record keeping. The BAMIC rehabilitation approach standardised the principles of rehabilitation while enabling an individualised rehabilitation programme, informed by previous work in this elite track and field group. That the medical staff were working full time within the sport, reporting within a performance team, gives confidence to the recording of the TRFT and identification of reinjury. However, as the BAMIC rehabilitation is intrinsically linked to the structural injury there is the undoubted limitation of a rehabilitation team that was not blinded to the MRI results resulting in risk of bias. This is particularly the case in the intratendon injuries for which the BAMIC approach advocates a 2-week delay in the progression of eccentric type exercise and accelerated running velocity to avoid placing elastic strain on the healing tendon. Thus, we cannot exclude that the MRI variables associated with longer TRFT are biased by the BAMIC rehabilitation protocol. This is mitigated somewhat as the performance culture within British Athletics demands appropriately expedited return to training and the coach and athlete were fully involved in shared decision-making regarding rehabilitation and performance targets. The TRFT relative to the existing literature suggests that rehabilitation outcomes were not inappropriately delayed due to cautious rehabilitation or prescribing rehabilitation progression based on imaging findings and TRFT from previously published studies.

What are the findings?

- British Athletics Muscle Injury Classification injury grade and intratendon class c injury are associated with increased time to return to full training in elite track and field.
- The cross-sectional area of tendon involvement and loss of tension are associated with time to return to full training rather than length of tendon involvement.

How might it impact on clinical practice in the future?

- The application of the British Athletics hamstring rehabilitation principles results in favourable time to return to full training with low reinjury rates.
- Hamstring injuries that involve the intramuscular tendon can be managed successfully with rehabilitation rather than surgery.

CONCLUSION

This study demonstrates the successful application of the British Athletics rehabilitation approach following BAMIC hamstring classification. The particularly low reinjury rates suggest that this targeted rehabilitation approach is helpful in successfully returning intratendon hamstring injuries to elite track and field. The key MRI variables are length and cross-section of muscle oedema, CSA of tendon involvement and loss of tension.

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